Software Requirements Specification — WebMsh

WebMsh — A Web-Based Mesh Generation Platform

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1. Product Overview

1.1 Purpose

This document specifies the software requirements for **WebMsh**, a web-based mesh generation platform built on top of the Gmsh Python API. It is intended to serve as the authoritative reference for the system's scope, features, constraints, and acceptance criteria throughout design, development, and testing.

WebMsh is a **server-side Gmsh instance with a web frontend**. The Python backend (FastAPI) runs the Gmsh API and exposes it via REST and WebSocket endpoints. The frontend (React + Three.js) provides a 3D viewer and interactive UI.

1.2 Intended Use and Target Audience

Intended use: WebMsh provides browser-based access to CAD geometry creation, manipulation, and finite element mesh generation — capabilities traditionally available only through desktop applications. Users interact with a 3D viewer and sidebar UI to build geometry, assign physical groups, generate meshes, and export results in standard simulation formats.

Target audience:

#	Audience	Description
1	**Engineering students**	Learning FEA/CFD workflows; need accessible meshing without installing native software.
2	**Academic researchers**	Preparing meshes for simulation codes (FEniCS, OpenFOAM, Code_Aster, Elmer).
3	**Mechanical / Civil / Aerospace engineers**	Building and meshing simple-to-moderate geometry for structural or thermal analysis.
4	**CFD practitioners**	Generating surface and volume meshes for fluid simulation.
5	**3D printing users**	Creating or importing geometry and exporting watertight STL meshes.

1.3 Product Vision Board

WebMsh Product Vision Board

VISION

What is your purpose for creating the product?

To make high-quality CAD-to-mesh workflows accessible from any modern browser, so users can create, inspect, and export simulation-ready meshes without desktop installation complexity.

What positive change should it bring about?

Reduce setup friction for students, researchers, and engineers; shorten time from geometry idea to solver-ready mesh; improve confidence through interactive 3D validation and clear error feedback; democratize meshing workflows for learning and prototyping.

Target group

Which market or market segments does the product address?

Engineering education, academic research groups, small-to-mid engineering teams, and 3D printing/prototyping users.

Who are the target customers and users?

Engineering students, researchers, CFD/FEA engineers, instructors, lab admins, and technical leads.

Needs

What problem does the product solve?

Complex local setup, fragmented CAD-to-mesh workflows, steep onboarding, and poor error visibility.

Which benefit does it provide?

Zero-install browser workflow, unified geometry-mesh-export process, faster learning curve, and clear operational feedback.

Product

What product is it?

WebMsh is a web-based mesh generation platform powered by FastAPI + Gmsh and React + Three.js for geometry creation/import, meshing, 3D inspection, and multi-format export.

What makes it stand out?

Browser-first access, integrated visual workflow, and interoperability for simulation toolchains.

Is it feasible to develop this product?

Yes. MVP feasibility is already proven with working geometry creation, file upload, meshing, rendering, and geometry management flows.

Business goals

How is the product going to benefit the organization?

Expands adoption among users who avoid heavy desktop tooling, strengthens education/research reach, and creates a base for advanced capabilities.

What are the business goals?

Improve activation, increase retention, maintain high reliability, drive recurring usage, and ensure high import/export success rates.

2. Planning

2.1 Product Initial Wish List

#	Wish List Item
1	Access the full meshing workflow from any modern browser without installing software.
2	Create basic 3D geometry (boxes, spheres, cylinders) through a visual interface rather than code.
3	Combine and modify shapes using boolean operations like union, cut, and intersect.
4	View geometry and mesh output interactively in a 3D workspace with orbit, pan, and zoom controls.
5	Import existing CAD files (STEP, STL) and have them render in the workspace immediately.
6	Generate finite element meshes with control over element size and algorithm.
7	Export meshes in formats compatible with common simulation tools (FEniCS, OpenFOAM, Abaqus).
8	Define named regions on the geometry (physical groups) by clicking directly in the 3D viewer.
9	Save work and return to it later — projects should persist across sessions.
10	Undo mistakes without losing the full model state.
11	Sign in securely and know that only you can access your projects and files.
12	See meaningful feedback when something goes wrong — not silent failures.
13	A UI that doesn't feel overwhelming — clean, focused, and task-driven.
14	A platform that can grow — room to add more primitives, advanced meshing, and solver integration later.

Key User Stories:

#	Role	User Story	Acceptance Condition
1	Engineering student	I want to create a box and generate a mesh so I can export it for my FEA assignment without installing Gmsh locally.	Box created, meshed, and exported as `.msh` entirely in browser.
2	CFD researcher	I want to import a STEP file of my geometry and generate a volume mesh so I can run it in OpenFOAM.	STEP file imports, surface renders correctly, 3D mesh generated and exported as `.vtk`.
3	General user	I want to orbit, pan, and zoom around my model so I can inspect it before meshing.	Camera controls work smoothly; geometry visible from all angles.
4	General user	I want to undo the last operation so I can recover from a bad boolean cut without starting over.	Undo restores previous model state correctly.
5	General user	I want to define a boundary region by clicking faces in the viewer so I can assign inlet/outlet conditions for simulation.	Physical group created with selected faces highlighted in distinct color.

#	Role	User Story	Acceptance Condition
6	General user	I want my project to save automatically so I don't lose work if my browser crashes.	Session restored with full geometry and mesh state on reconnect.
7	General user	I want clear error messages when geometry or meshing fails so I know what to fix.	Descriptive toast notification shown; model state unchanged after failure.
8	System admin	I want user sessions to be authenticated and time-limited so that the server isn't exposed to unauthorized use.	JWT tokens expire after 1 hour; refresh token valid for 7 days; unauthenticated requests rejected.

2.2 Sprint List

Sprint	Title	Outcome
1	Project Setup and Core Architecture	A running two-part web system with frontend-backend communication confirmed, basic API routes live, and the development environment fully established.
2	Primitive Geometry Creation and Mesh Generation	End-to-end geometry creation working — users can create boxes, spheres, and cylinders via the API and receive valid mesh data back.
3	Frontend 3D Viewer and Interaction	A functional visual workspace where users interact with geometry through forms, see results in a Three.js viewport, and control the camera.
4	CAD/Mesh File Import and Geometry Management	File-based workflows added — users can upload STEP and STL files, view imported geometry, delete objects, and receive action feedback.

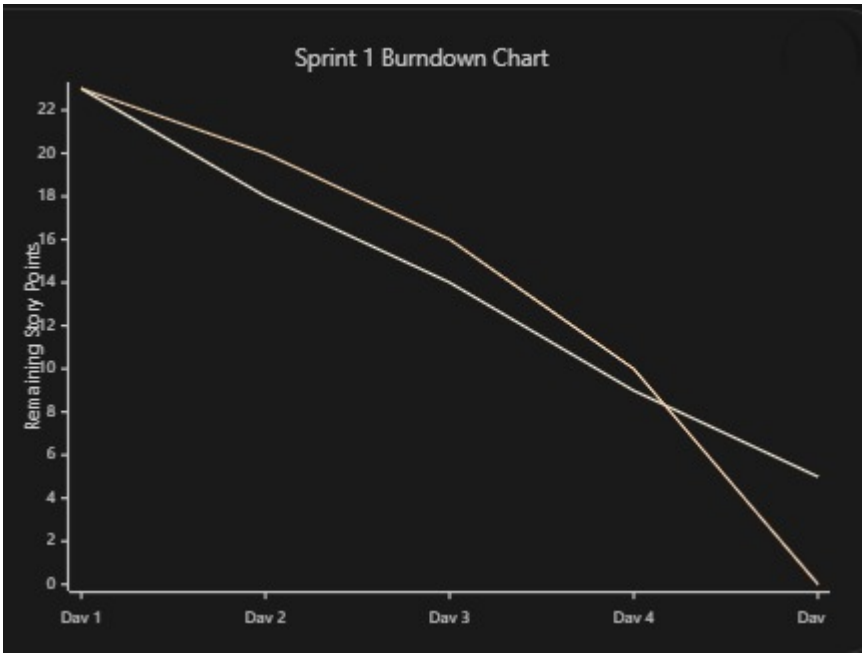
2.3 Tasks Per Sprint

Sprint 1: Project Setup and Core Architecture — Total SP: 23

#	Task	SP
1	Create the top-level repository structure for `frontend` and `backend`.	1
2	Initialize the frontend using Vite and React.	2
3	Initialize the backend using FastAPI and Uvicorn.	2
4	Add backend dependency definitions in `requirements.txt`.	1
5	Add frontend dependency definitions in `package.json`.	1
6	Configure ESLint for frontend code quality.	2
7	Configure Vite for frontend builds.	1

#	Task	SP
8	Create the backend application instance in `main.py`.	1
9	Add a root route for confirming API availability.	1
10	Add a health-check route for service monitoring.	1
11	Add an info route to expose app name, version, Gmsh availability, and geometry count.	2
12	Add CORS middleware for frontend-backend communication.	2
13	Verify local development connectivity between frontend and backend.	3
14	Define system scope, constraints, and initial feature requirements.	3

Sprint 1 Burndown Chart

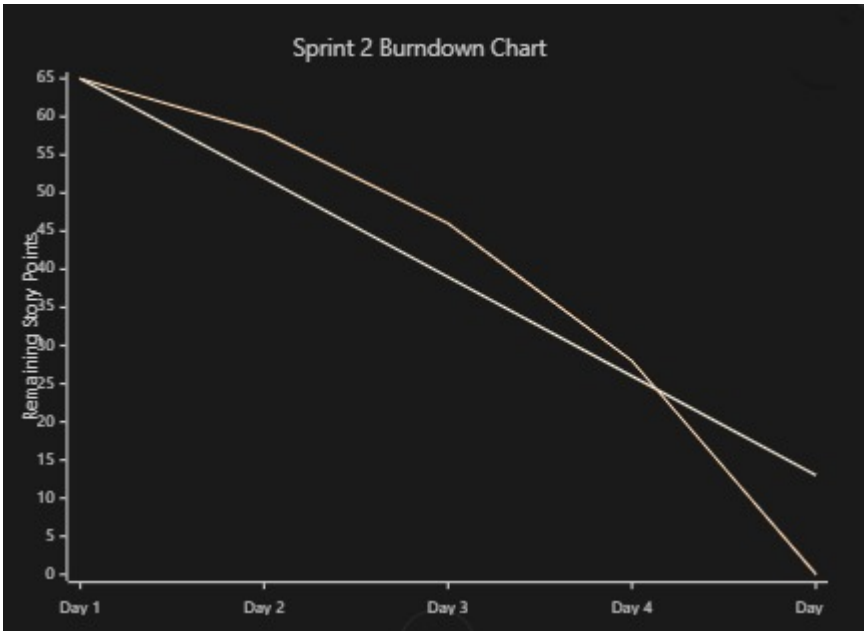


Sprint 2: Primitive Geometry Creation and Mesh Generation — Total SP: 65

#	Task	SP
1	Define `BoxRequest`, `SphereRequest`, and `CylinderRequest` models.	3
2	Define `MeshNode`, `Mesh`, and `Geometry` response models.	3
3	Implement `/geometry/box` endpoint.	5
4	Implement `/geometry/sphere` endpoint.	5
5	Implement `/geometry/cylinder` endpoint.	5
6	Add validation rules to ensure positive dimensions.	2
7	Integrate Gmsh initialization and cleanup inside each geometry workflow.	5
8	Implement `_build_box_mesh()` for box meshing.	5

#	Task	SP
9	Implement `_build_sphere_mesh()` for sphere meshing.	5
10	Implement `_build_cylinder_mesh()` for cylinder meshing.	5
11	Implement `_extract_surface_mesh()` to normalize mesh output.	8
12	Convert Gmsh node and element data into API-friendly JSON structures.	5
13	Store generated geometry in an in-memory list.	2
14	Implement `/geometry` GET endpoint to list all geometry objects.	2
15	Test that primitive creation returns valid node and triangle data.	5

Sprint 2 Burndown Chart

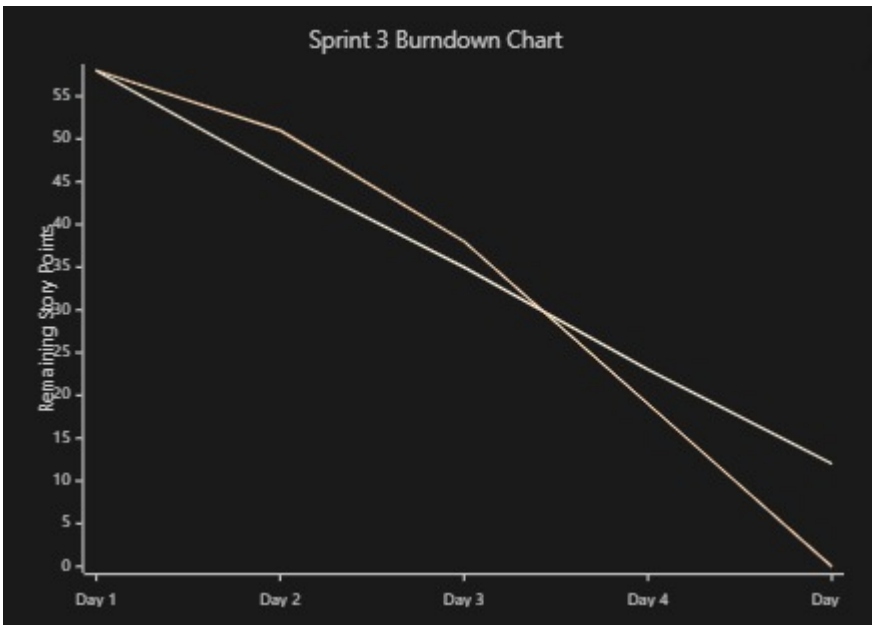


Sprint 3: Frontend 3D Viewer and Interaction — Total SP: 58

#	Task	SP
1	Create the main React application component.	2
2	Initialize a Three.js scene on component mount.	5
3	Create perspective camera and WebGL renderer.	3
4	Add ambient and directional lighting.	1
5	Add grid helper and axes helper for scene orientation.	1
6	Add `OrbitControls` for user interaction.	3
7	Build animation/render loop.	3
8	Add responsive resize handling for the 3D canvas.	2

#	Task	SP
9	Create a geometry group container for rendered objects.	2
10	Build frontend logic to translate API mesh data into Three.js geometry.	8
11	Add wireframe/edge rendering for box geometry.	3
12	Add wireframe/edge rendering for sphere geometry.	3
13	Add wireframe/edge rendering for cylinder geometry.	3
14	Build forms for entering box dimensions and origin.	3
15	Build forms for entering sphere radius and center.	3
16	Build forms for entering cylinder dimensions and base position.	3
17	Connect form actions to backend API methods.	5
18	Load health, system info, and existing geometry on app startup.	3
19	Display backend status and geometry count in the sidebar.	2
20	Add sidebar collapse and expand behavior.	3

Sprint 3 Burndown Chart

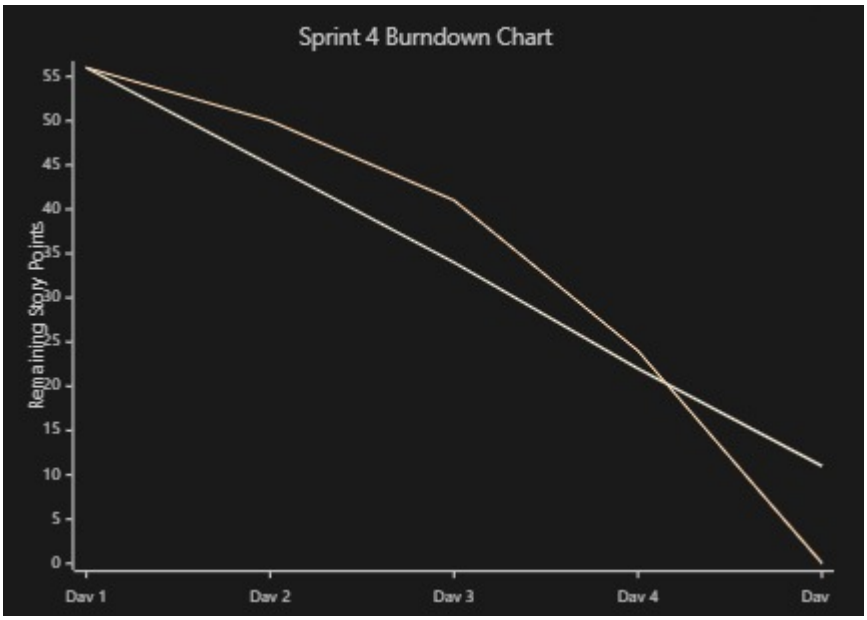


Sprint 4: CAD/Mesh File Import and Geometry Management — Total SP: 56

#	Task	SP
1	Implement `/geometry/upload` endpoint for multipart file upload.	5
2	Validate supported input file formats.	2
3	Store uploaded files in temporary backend storage.	3

#	Task	SP
4	Implement `_build_mesh_from_cad()` to open uploaded files through Gmsh.	8
5	Detect whether uploaded files already contain mesh data.	5
6	Generate a surface mesh if one is not already present.	5
7	Clean up temporary files after processing.	2
8	Return uploaded geometry as a normal geometry object with mesh data.	3
9	Add file input support in the frontend.	2
10	Add upload action button and wire it to the API.	2
11	Add API helper for file upload using `FormData`.	2
12	Render uploaded CAD/mesh results in the 3D scene.	5
13	Implement geometry deletion endpoint in the backend.	2
14	Implement delete action in the frontend API layer.	2
15	Add delete buttons for each geometry entry in the sidebar.	2
16	Refresh geometry state after create, upload, and delete operations.	3
17	Add user-facing action messages for success and failure cases.	2

Sprint 4 Burndown Chart



Grand Total: 202 Story Points

2.4 Sprint Journal

Role Rotation Overview

Sprint	Duration	My Role
Sprint 1	Feb 10 – Feb 14, 2025	Product Owner
Sprint 2	Feb 17 – Feb 21, 2025	Scrum Master
Sprint 3	Feb 23 – Feb 27, 2025	Developer 1
Sprint 4	March 1 – March 5, 2025	Developer 2

Sprint 1 — Project Setup & Core Architecture

Duration: Feb 10 – Feb 14, 2025
My Role: Product Owner

> **Sprint Goal:** Establish project foundation, define SRS expectations, and ensure backend–frontend setup aligns with requirements.

Daily Journal

Day 1 — Monday, Feb 10

Defined initial SRS requirements and acceptance criteria for core backend endpoints.

Day 2 — Tuesday, Feb 11

Reviewed SRS Section 1 product wishlist items and aligned them with Sprint 1 deliverables.

Day 3 — Wednesday, Feb 12

Validated `/health` endpoint response matches SRS expectations.

Day 4 — Thursday, Feb 13

Drafted acceptance tests for CORS and route availability.

Day 5 — Friday, Feb 14

Accepted Sprint 1 deliverables: repository structure, backend startup, and endpoint readiness.

Sprint 1 Reflection

> *As Product Owner in Sprint 1, I learned the importance of clearly defining acceptance criteria early. Ensuring alignment with the SRS helped avoid ambiguity during development.*

Key learning: Strong acceptance criteria early prevents confusion later.

Sprint 1 Retrospective

Category	Notes
What went well	Clear SRS alignment ensured smooth validation of endpoints.
What didn't go well	Some acceptance criteria needed refinement mid-sprint.
What I'll improve	Define more precise and testable criteria upfront.

Sprint 2 — Primitive Geometry Creation & Mesh Generation

Duration: Feb 17 – Feb 21, 2025
My Role: Scrum Master

> **Sprint Goal:** Ensure smooth execution of backend mesh generation tasks and unblock dependencies like Gmsh.

Daily Journal

Day 6 — Monday, Feb 17

Opened Sprint 2 planning. Identified Gmsh availability as a potential blocker.

Day 7 — Tuesday, Feb 18

Gmsh blocker cleared by Developer 1. Reminded team to proceed with mesh integration.

Day 8 — Wednesday, Feb 19

Checked burn-down: 8/15 tasks completed — sprint on track.

Day 9 — Thursday, Feb 20

Tracked 13/15 tasks completed. Scheduled Sprint 2 review.

Day 10 — Friday, Feb 21

Facilitated Sprint 2 retrospective. Noted dependency risks and team coordination improvements.

Sprint 2 Reflection

> *Taking on the Scrum Master role after being Product Owner gave me a broader perspective on execution. Managing blockers like Gmsh taught me the importance of early risk identification.*

Key learning: Identifying blockers early keeps the sprint on track.

Sprint 2 Retrospective

Category	Notes
What went well	Strong tracking of sprint progress and quick resolution of Gmsh blocker.
What didn't go well	Initial dependency risk was not flagged early enough during planning.
What I'll improve	Proactively identify and escalate risks before sprint start.

Sprint 3 — Frontend 3D Viewer & Interaction

Duration: Feb 23 – Feb 27, 2025
My Role: Developer 1

> **Sprint Goal:** Build core 3D viewer functionality and integrate frontend with backend mesh data.

Daily Journal

Day 11 — Monday, Feb 23

Initialized Three.js scene inside React application.

Day 12 — Tuesday, Feb 24

Completed lighting setup (ambient + directional lighting).

Day 13 — Wednesday, Feb 25

Converted API mesh data (nodes and triangles) into Three.js geometry.

Day 14 — Thursday, Feb 26

Built parameter input forms for box, sphere, and cylinder.

Day 15 — Friday, Feb 27

Completed startup fetch for `/health` and existing geometry list.

Sprint 3 Reflection

> *Coding as Developer 1 gave me deep appreciation for how frontend rendering depends on clean backend data structures. Converting mesh data into visuals was a key challenge.*

Key learning: Clean backend data design simplifies frontend implementation.

Sprint 3 Retrospective

Category	Notes
What went well	Successful integration of mesh data into 3D rendering pipeline.
What didn't go well	Initial data conversion took longer due to format understanding.
What I'll improve	Review data contracts earlier before implementation.

Sprint 4 — CAD/Mesh File Import & Geometry Management

Duration: March 1 – March 5, 2025
My Role: Developer 2

> **Sprint Goal:** Implement file upload, CAD mesh rendering, and geometry management features.

Daily Journal

Day 16 — Monday, March 1

Assigned frontend file input and upload wiring.

Day 17 — Tuesday, March 2

Completed file input and wired FormData submission to backend.

Day 18 — Wednesday, March 3

Uploaded CAD mesh successfully renders in 3D viewer.

Day 19 — Thursday, March 4

Added per-item delete buttons in sidebar geometry list.

Day 20 — Friday, March 5

Implemented user-facing success/failure action messages.

Sprint 4 Reflection

> *As Developer 2 in the final sprint, I handled user-facing features. Seeing file uploads and mesh rendering work end-to-end was very satisfying.*

Key learning: End-to-end features require tight frontend–backend coordination.

Sprint 4 Retrospective

Category	Notes
What went well	Successful implementation of file upload and geometry management features.
What didn't go well	Minor UI refinements required late in the sprint.
What I'll improve	Allocate time for UI polishing before sprint end.

Overall Reflection

Across all four sprints, I transitioned through multiple roles, gaining both technical and leadership experience:

Sprint	Role	My Biggest Contribution
Sprint 1	Product Owner	Defined acceptance criteria and validated SRS alignment
Sprint 2	Scrum Master	Managed sprint execution and resolved Gmsh blocker
Sprint 3	Developer 1	Built 3D viewer and mesh rendering pipeline
Sprint 4	Developer 2	Implemented file upload and geometry management

This rotation helped me understand the full lifecycle of a Scrum project—from planning and coordination to implementation and delivery.

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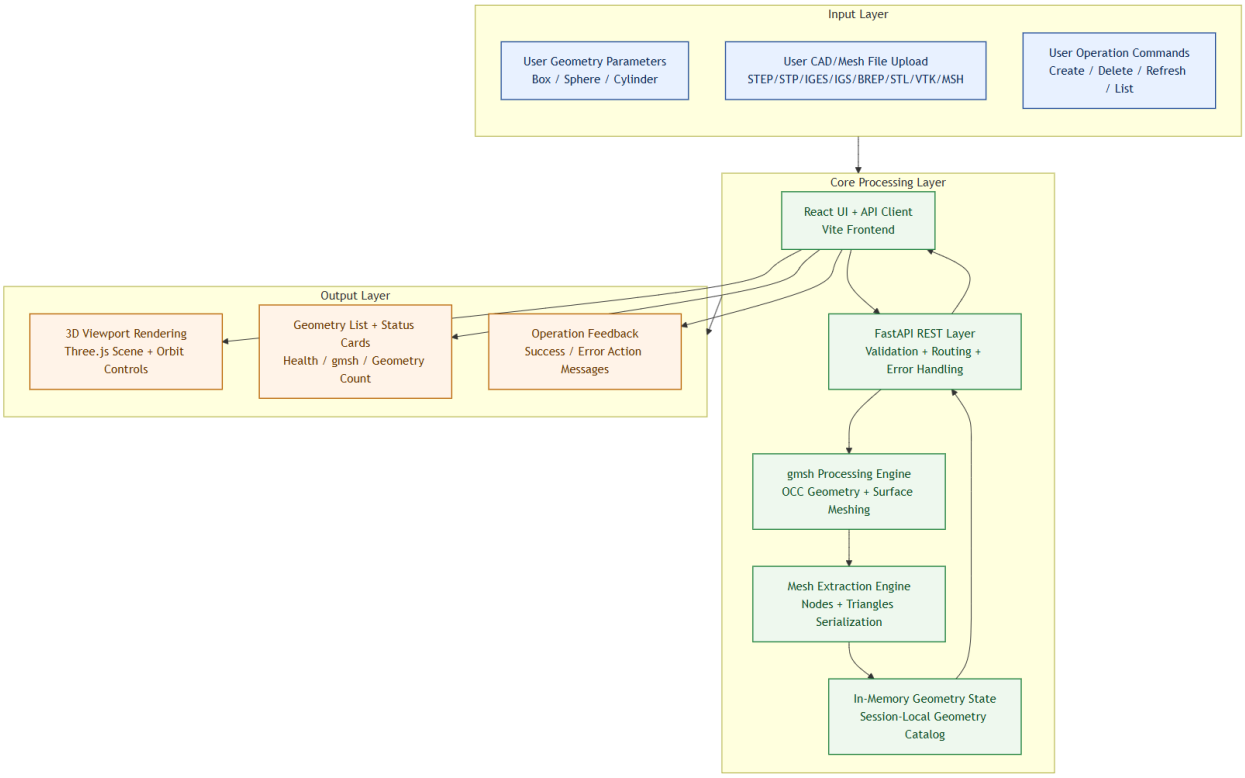
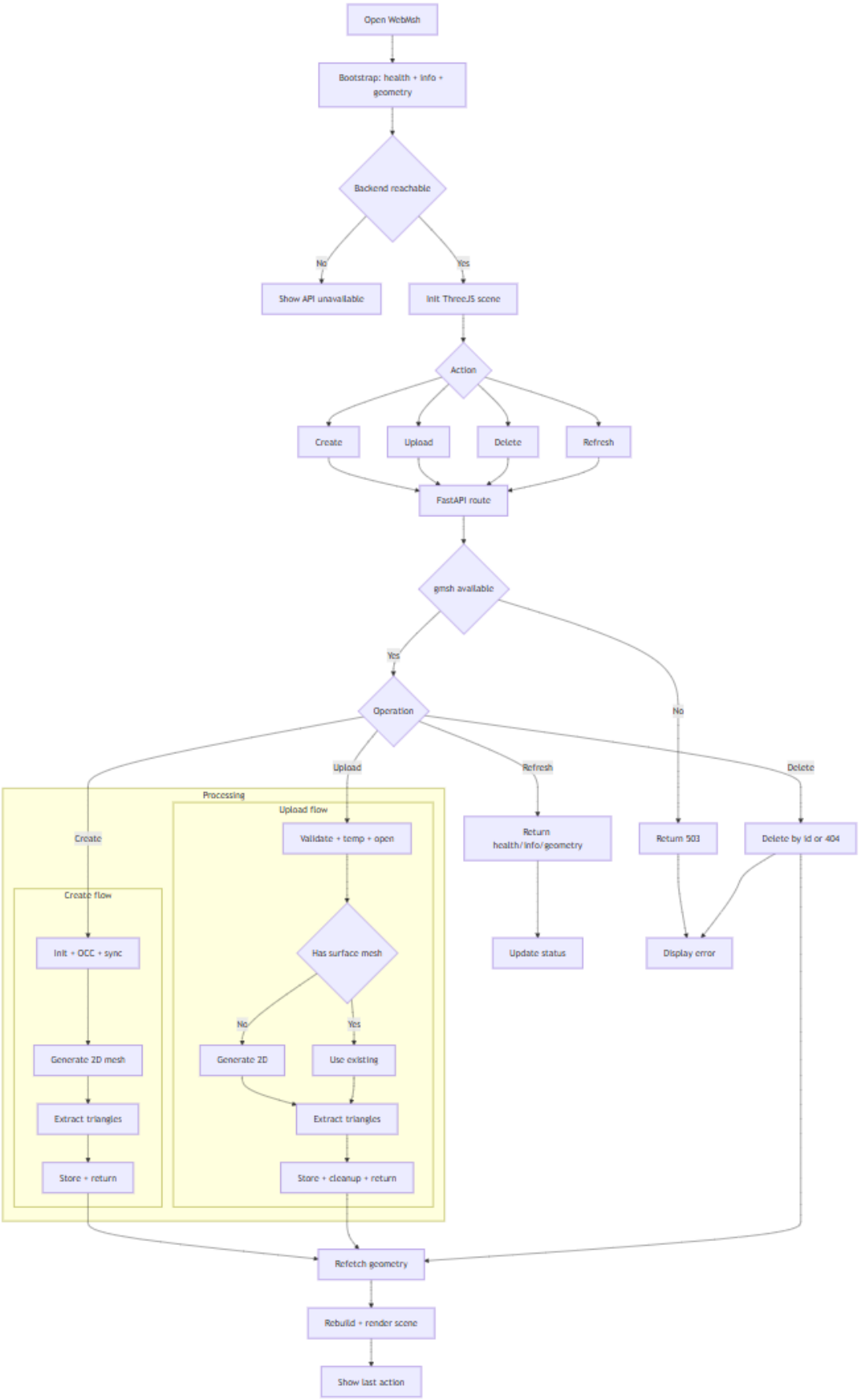
3. Workflows, UML Diagram and Wireframes

3.1 Workflows

3.1.1 High-Level Design Workflow (Input -> Core -> Output)

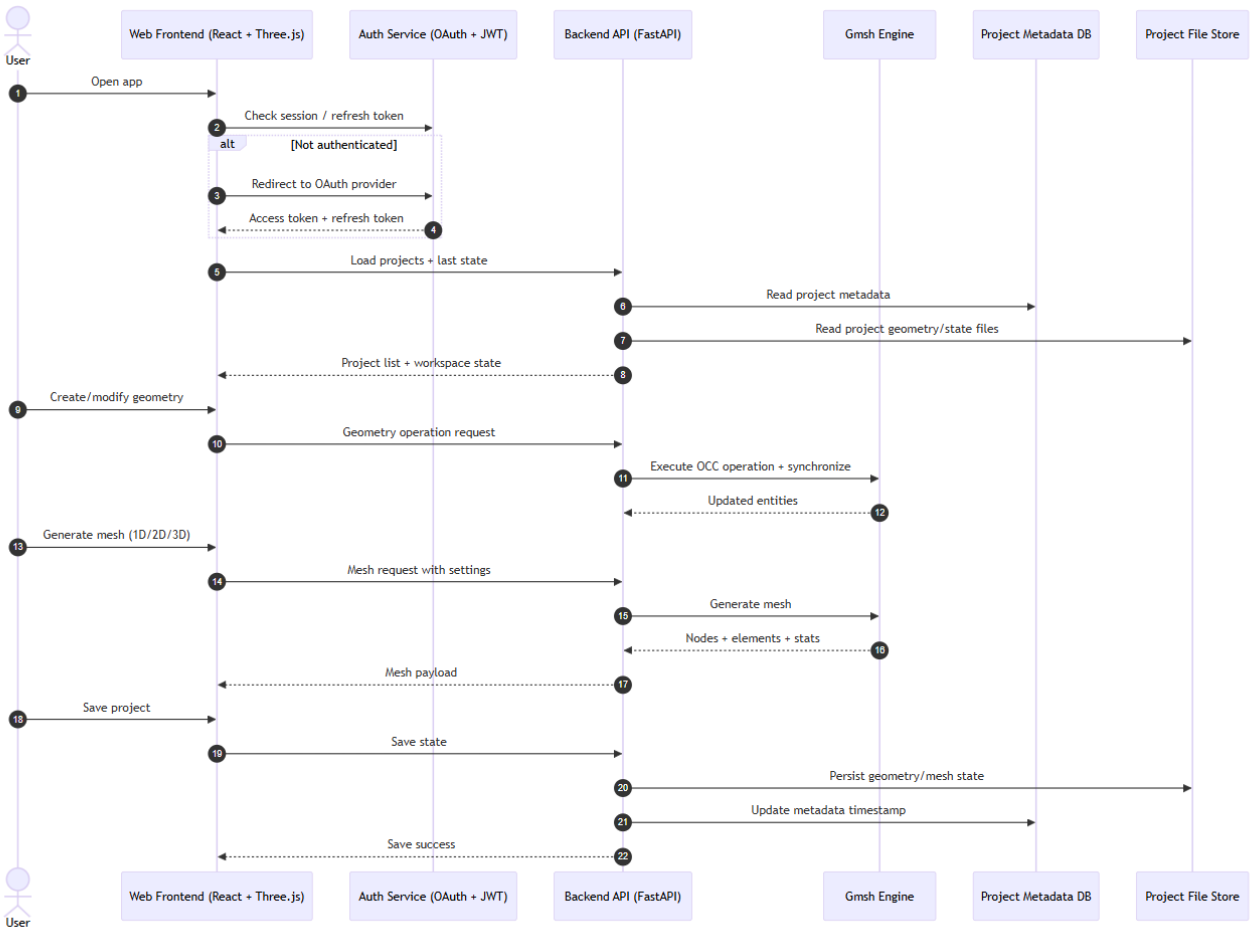


3.1.2 Detailed Workflow Diagram (Operational Flow)

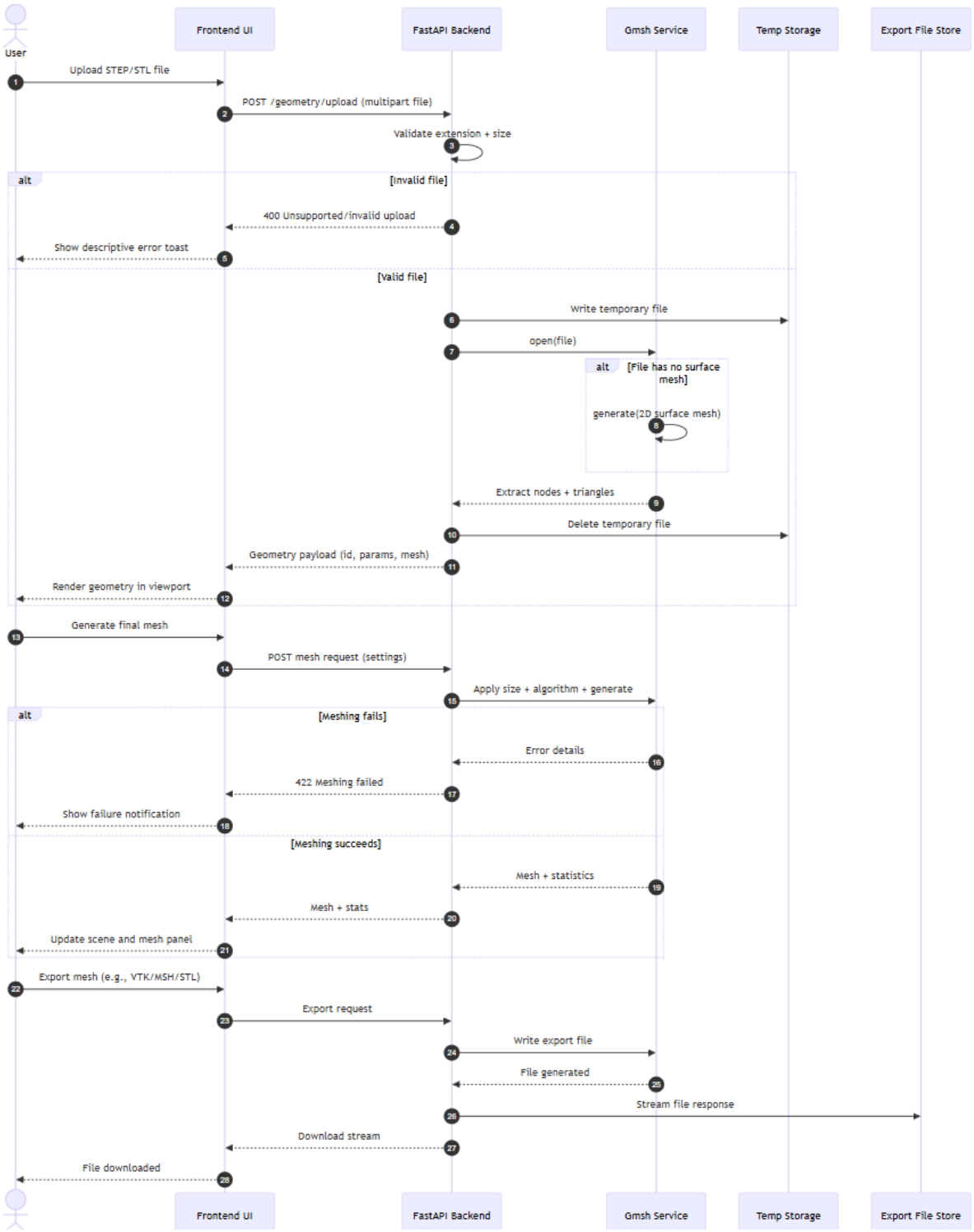


3.2 UML Diagrams

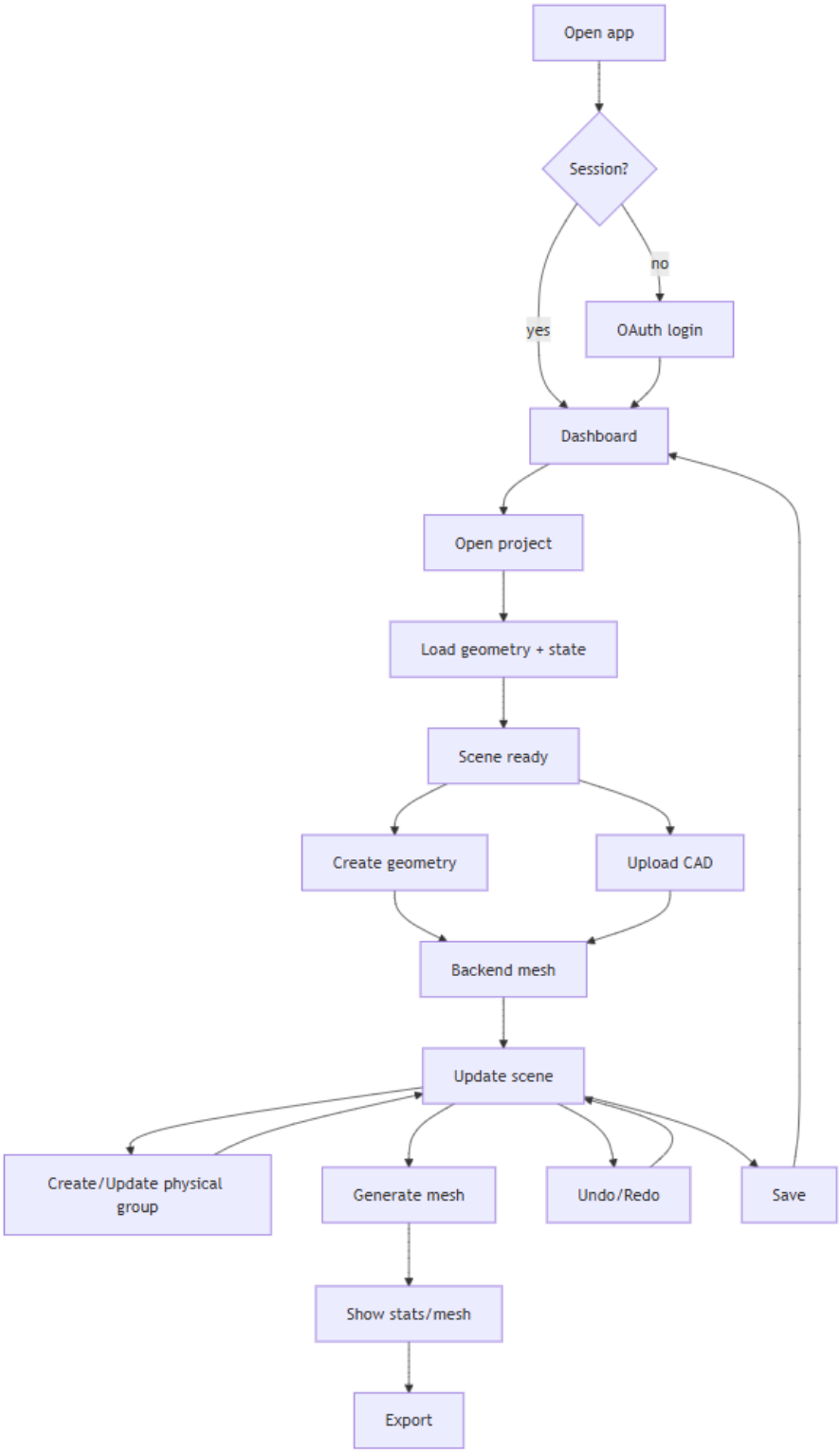
3.2.1 Interaction Diagram



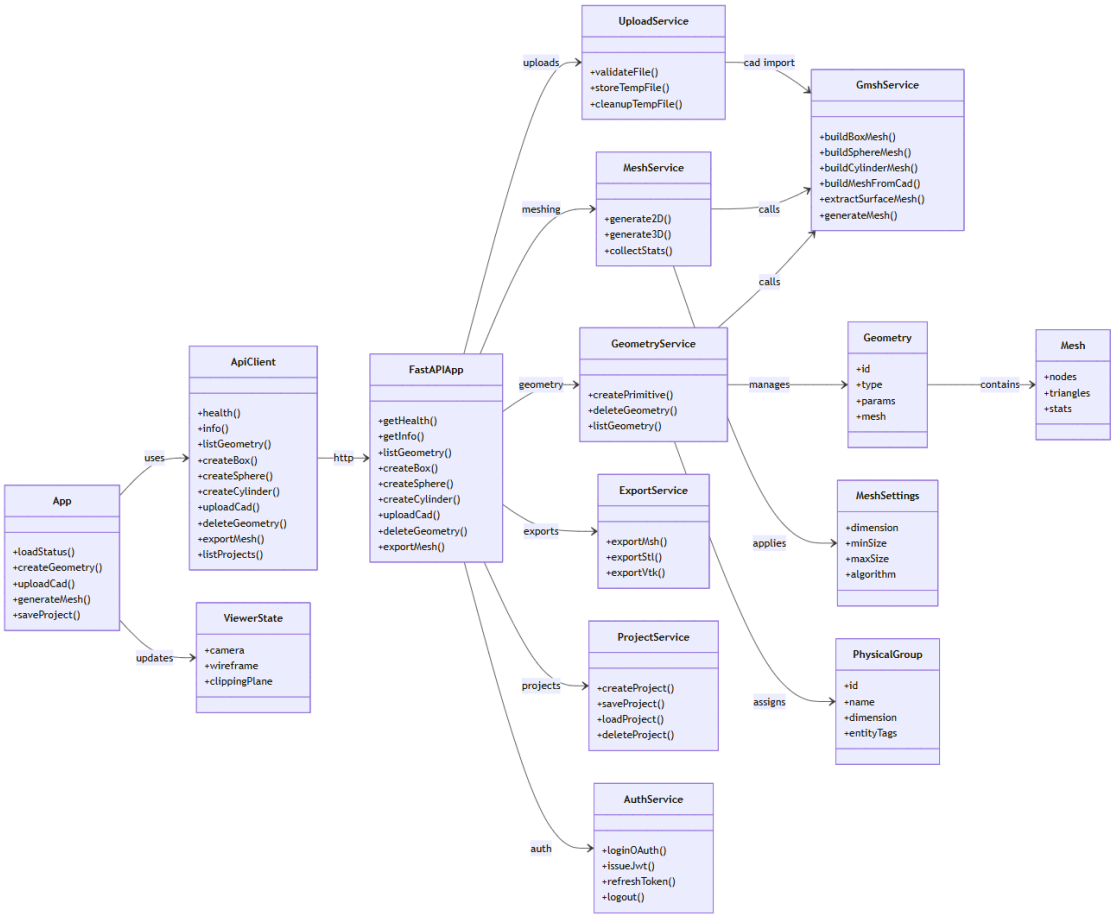
3.2.2 Sequence Diagram



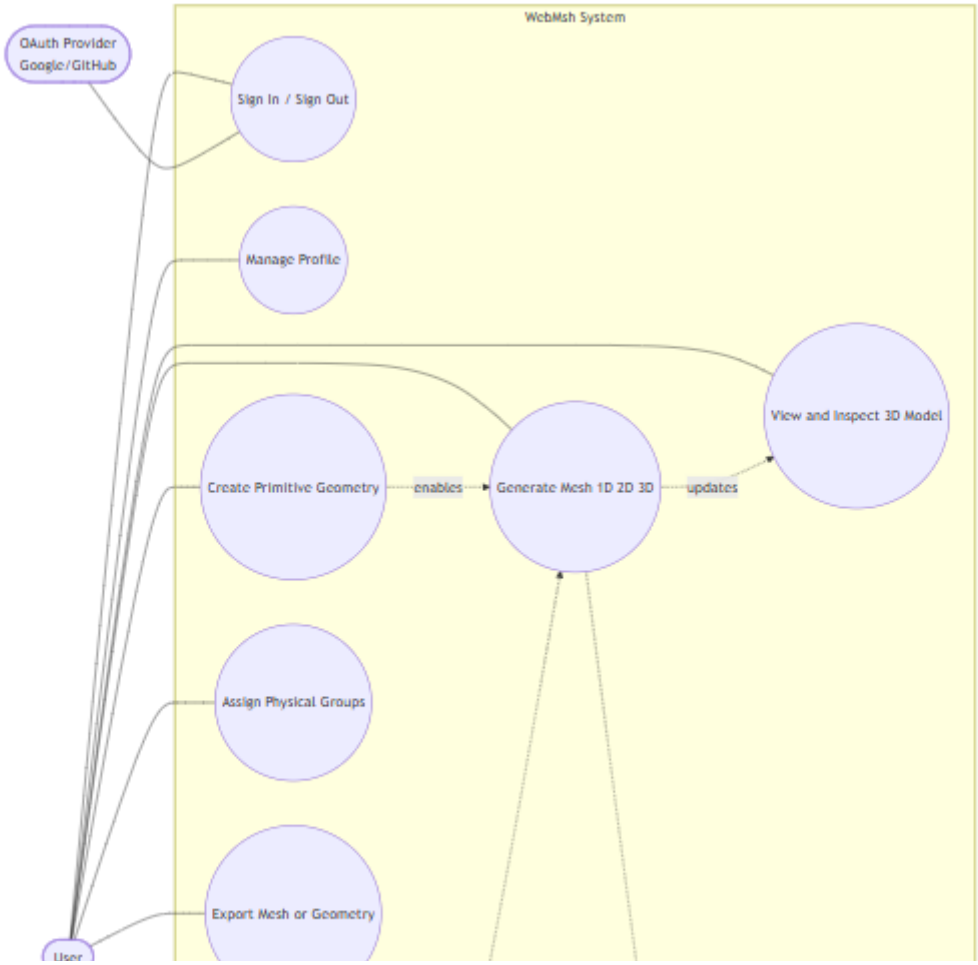
3.2.3 Activity Diagram

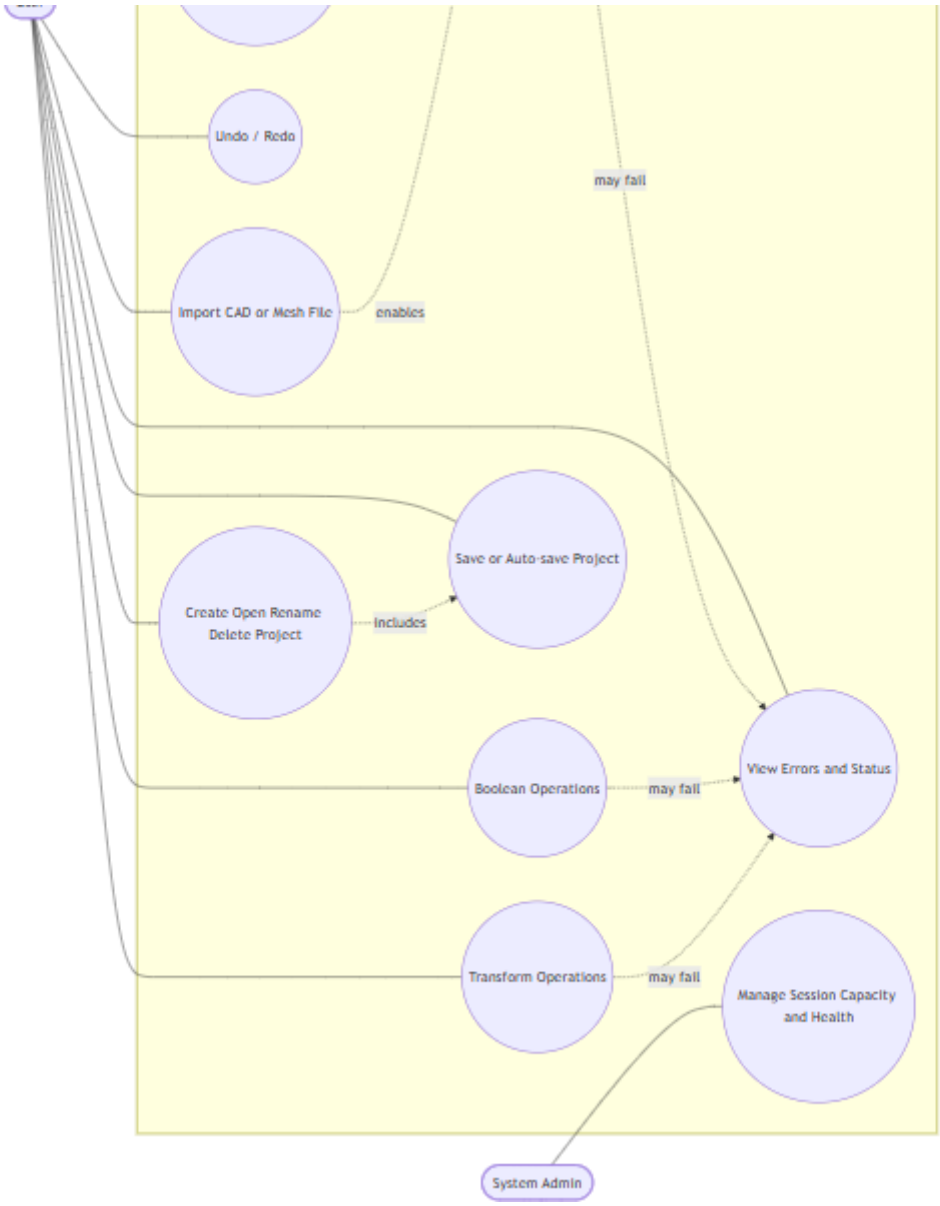


3.2.4 Class Diagram



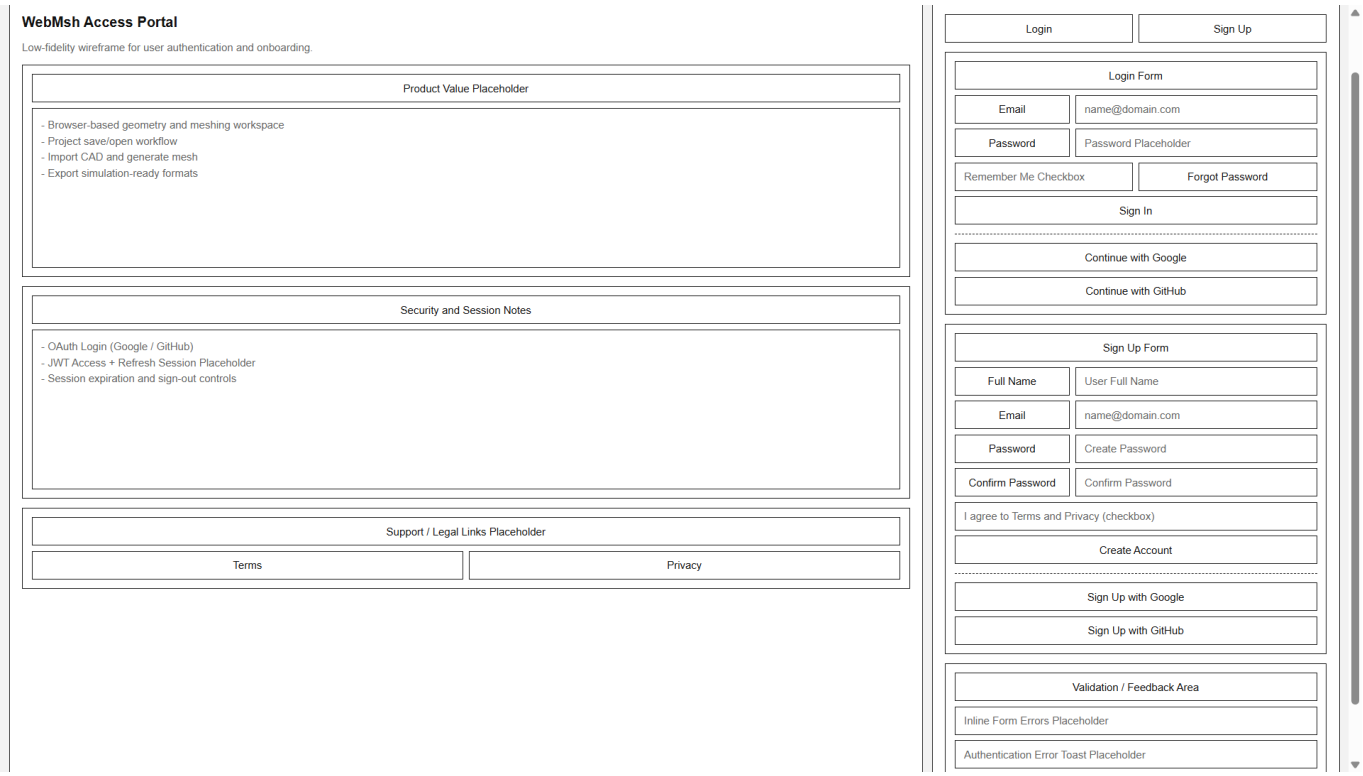
3.2.5 Use Case Diagram



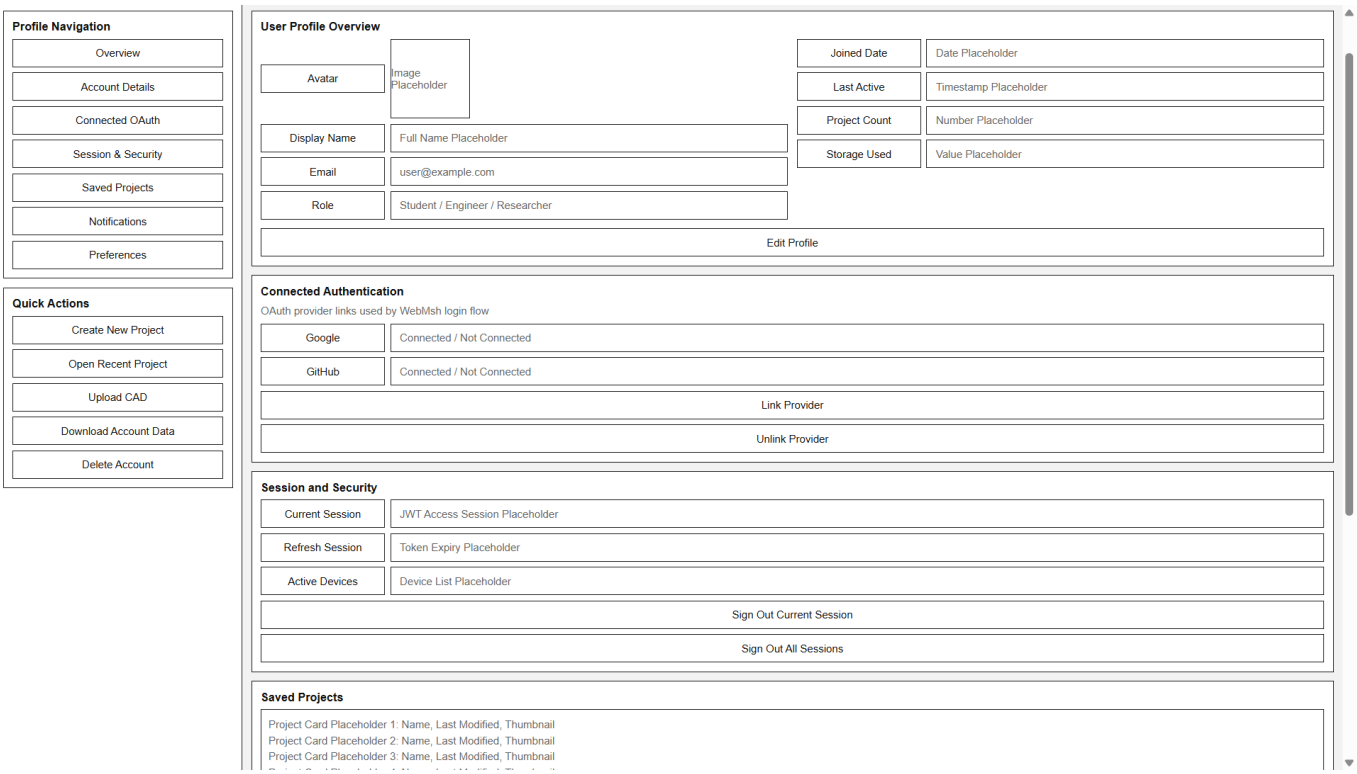


3.3 Wireframes

3.3.1 User Authenication Wireframe



3.3.2 User Profile Wireframe



3.3.3 Main Application Wireframe

4. Features and Requirements

4.1 Features

4.1.1 Authentication and User Management

#	Feature
1	OAuth sign-in via Google or GitHub.
2	JWT-based session tokens with expiration and refresh.
3	Sign-out and session invalidation.
4	User profile display (name, avatar from OAuth provider).

4.1.2 Project Management

#	Feature
1	Create a new project with a name.
2	List all projects for the authenticated user (name, last modified date, thumbnail).
3	Open an existing project and restore its full geometry/mesh state.
4	Rename a project.
5	Delete a project.
6	Manual save via Ctrl+S or toolbar button.

4.1.3 Geometry Creation

#	Feature	Gmsh API
1	Create a Box (origin, dx, dy, dz).	`occ.addBox()`
2	Create a Sphere (center, radius).	`occ.addSphere()`
3	Create a Cylinder (center, axis, radius, length).	`occ.addCylinder()`
4	Create a Cone (center, axis, r1, r2, length).	`occ.addCone()`
5	Create a Torus (center, r1, r2).	`occ.addTorus()`
6	Extrude a surface along a direction vector.	`occ.extrude()`
7	Revolve a surface around an axis by a specified angle.	`occ.revolve()`

4.1.4 Boolean Operations

#	Feature	Gmsh API
1	Union (fuse) two or more volumes.	`occ.fuse()`
2	Cut (subtract) one volume from another.	`occ.cut()`
3	Intersect two or more volumes.	`occ.intersect()`

4.1.5 Geometry Transforms

#	Feature	Gmsh API
1	Translate entities by a vector (dx, dy, dz).	`occ.translate()`
2	Rotate entities around an axis by an angle.	`occ.rotate()`
3	Scale entities from a center point by a factor.	`occ.dilate()`
4	Copy entities.	`occ.copy()`
5	Delete selected entities.	`occ.remove()`

4.1.6 Geometry Import

#	Feature
1	Upload and import STEP files (.step, .stp) via file picker or drag-and-drop.
2	Upload and import STL files (.stl) via file picker or drag-and-drop.

4.1.7 Physical Groups

#	Feature

#	Feature
1	Create a physical group for a given dimension (0D, 1D, 2D, 3D).
2	Assign a name to a physical group.
3	Add selected entities to a physical group via click-to-select in the 3D viewer.
4	Remove entities from a physical group.
5	Delete a physical group.
6	Color-code physical groups in the 3D viewer with distinct colors.

4.1.8 Export

#	Feature	Format
1	Export mesh in Gmsh native format.	`.msh`
2	Export mesh as STL.	`.stl`
3	Export mesh in VTK format.	`.vtk`
4	Export geometry as STEP.	`.step`
5	Export mesh in Abaqus format.	`.inp`
6	Export mesh in Ideas/Universal format.	`.unv`

4.1.9 Workflow and UX

#	Feature
1	Model tree / entity browser sidebar showing all entities organized by dimension.
2	Entity property inspector: clicking an entity displays its bounding box, area or volume, and centroid.
3	Undo and redo via Ctrl+Z / Ctrl+Y or toolbar buttons.
4	Loading indicators during mesh generation and geometry operations.
5	Error notifications (toast messages) for failed operations with descriptive messages.
6	Keyboard shortcuts: Delete (remove entity), Ctrl+Z (undo), Ctrl+Y (redo), Escape (deselect).

4.2 Functional Requirements

4.2.1 Authentication

#	Requirement
1	The system shall redirect unauthenticated users to the sign-in page.
2	The system shall authenticate users via OAuth 2.0 Authorization Code flow with a supported provider (Google or GitHub).

#	Requirement
3	Upon successful OAuth callback, the system shall issue a JWT access token (1-hour expiry) and a refresh token (7-day expiry).
4	The system shall reject API requests with expired or invalid JWT tokens with HTTP 401.
5	The system shall store user records (OAuth provider ID, display name, email, avatar URL) in the database upon first sign-in.

4.2.2 Gmsh Session Management

#	Requirement
1	The system shall allocate one Gmsh process per active user session.
2	Each Gmsh process shall be isolated: operations in one session shall not affect another.
3	The system shall terminate idle Gmsh processes after 30 minutes of inactivity.
4	The system shall limit the maximum number of concurrent Gmsh sessions based on server configuration (default: 20).
5	If maximum sessions are reached, new session requests shall receive HTTP 503 with a "server at capacity" message.

4.2.3 Geometry Operations

#	Requirement
1	The system shall support creation of geometry primitives and return updated model entities after each operation.
2	Boolean operations shall accept object and tool entities and return the resulting geometry.
3	If a geometry operation fails, the system shall return a descriptive error and leave the model state unchanged.
4	File import shall accept STEP and STL uploads (max 50 MB) and merge them into the current model.

4.2.4 Mesh Operations

#	Requirement
1	Mesh generation shall support 1D, 2D, and 3D dimensions and apply configured mesh settings before generating.
2	Mesh statistics shall report node count, element count by type, and min/max element size.
3	Mesh export shall return the file as a downloadable stream in the selected format.
4	If mesh generation fails, the system shall return a descriptive error message.

4.2.5 Physical Groups

#	Requirement
1	Physical groups shall be persisted as part of the project save state.
2	The system shall assign a unique color to each physical group automatically. Colors shall be distinct and visually differentiable (up to 12 groups; beyond 12, colors may repeat).
3	The system shall return entity-to-physical-group mappings so the frontend can apply per-face coloring.

4.3 External Interface Requirements

4.3.1 User Interface

#	Requirement
1	The application shall present a single-page layout with: (a) a left sidebar for tools, forms, and the model tree; (b) a central 3D viewport; (c) a top toolbar for global actions (save, undo, redo, export, theme toggle).
2	The sidebar shall be collapsible to maximize viewport area.
3	Geometry creation forms shall validate input (e.g., radius > 0, dimensions > 0) and display inline validation errors before submission.
4	The 3D viewport shall occupy at least 60% of the screen width at all times.
5	The sign-in page shall display the application name, a brief description, and an OAuth sign-in button.
6	The project dashboard shall display projects as cards with name, last-modified timestamp, and a thumbnail preview.

4.3.2 Backend API

#	Requirement
1	The backend shall expose a RESTful API over HTTPS on a configurable port (default: 8000).
2	All API endpoints (except `/auth/*`) shall require a valid JWT in the `Authorization: Bearer <token>` header.
3	The API shall return JSON for metadata responses and binary buffers for mesh/geometry data.
4	The API shall provide OpenAPI 3.0 documentation at `/docs` (auto-generated by FastAPI).
5	A WebSocket endpoint (`/ws/session/{session_id}`) shall be available for real-time progress updates during long-running operations (mesh generation).

4.3.3 External Systems

#	Requirement
1	The system shall integrate with Google OAuth 2.0 or GitHub OAuth for user authentication.

#	Requirement
2	The system shall use PostgreSQL (production) or SQLite (development) for persistent storage of user accounts and project metadata.
3	The system shall use the server filesystem for storing project geometry files (BREP) and exported meshes.

4.4 Non-Functional Requirements

4.4.1 Security

#	Requirement
1	All client-server communication shall be encrypted via TLS (HTTPS / WSS).
2	JWT tokens shall be stored in HTTP-only, Secure, SameSite cookies — not localStorage.
3	File upload shall validate file extension and MIME type, and reject files exceeding the 50 MB size limit.
4	User-uploaded files shall be stored in sandboxed directories, isolated by user ID. A user shall not be able to access another user's files.
5	Server-side Gmsh processes shall not have network access or filesystem access beyond their designated project directory.
6	The API shall enforce rate limiting: maximum **100 requests per minute** per authenticated user.

4.4.2 Reliability

#	Requirement
1	Auto-save shall ensure that no more than **30 seconds** of work is lost in the event of a server crash or browser close.
2	The system shall gracefully handle browser disconnection: the Gmsh session persists for 30 minutes, and the user can reconnect and resume.

4.5 Assumptions and Dependencies

#	Assumption / Dependency
1	**Gmsh Python package** (`pip install gmsh`) is available and functional on the deployment platform (Linux x86_64).
2	**Python 3.10+** is available on the server.
3	**OpenCASCADE** is bundled with the Gmsh package (this is the default for the PyPI distribution).
4	Users have a modern browser with **WebGL 2.0** support (Chrome 56+, Firefox 51+, Edge 79+, Safari 15+).

#	Assumption / Dependency
5	Users authenticate via a third-party OAuth provider (Google or GitHub). The OAuth provider's API is assumed to be available.
6	A **PostgreSQL** (or SQLite for development) database is available for user accounts and project metadata.
7	Server deployment infrastructure supports **Docker** and provides at least 2 GB RAM per concurrent user session.
8	Project geometry state can be reliably serialized to and deserialized from BREP format + JSON metadata.

4.6 Languages and Tools Used

4.6.1 Backend

Tool / Language	Purpose
Python 3.10+	Primary backend language
Uvicorn	ASGI server hosting the FastAPI application
FastAPI	REST API framework, request routing, OpenAPI documentation
Pydantic	Request/response validation and schema definition
Python-multipart	Multipart form data parsing for file uploads
Gmsh Python API	Geometry creation, mesh generation, CAD operations
OpenCASCADE	CAD kernel bundled with Gmsh for geometry processing

4.6.2 Frontend

Tool / Language	Purpose
JavaScript / React	Primary frontend language and UI framework
Vite	Build tool with fast HMR and JSX transformation
Three.js	WebGL 2.0 3D rendering and viewer
CSS	Layout, theming, and responsive design
ESLint	Code quality and linting
TypeScript	Type definitions for IDE support

4.6.3 Database

Tool	Purpose
PostgreSQL	Production database for user accounts and project metadata

Tool	Purpose
SQLite	Development database

4.6.4 Auth and Communication

Tool	Purpose
JWT	Session token authentication
Google OAuth 2.0 / GitHub OAuth	Third-party user authentication
WebSocket	Real-time progress updates during long-running operations